

## Problem 7: Uniform Lattices with 2-Torsion as Fundamental Groups

**Problem.** Q7. Contributor field: Shmuel Weinberger (University of Chicago).

**Theorem 1** (RMA generated solution, iteration 1). *No. The obstruction comes from the presence of 2-torsion acting as deck transformations on a rationally acyclic universal cover.*

*Proof. Parsing of the statement.* The problem is treated as a research problem in lattices, transformation groups, and topology. The quantifier structure used in the proof is:

- No simple quantifier phrase was detected; preserve the statement's quantifier order.

The definitions extracted from the statement are:

- No explicit 'let ...' definition was detected; use the problem statement verbatim.

**Answer and construction.** No. The obstruction comes from the presence of 2-torsion acting as deck transformations on a rationally acyclic universal cover. The object or method used to witness the answer is the following: Assume such a compact manifold exists and let an element of order two in  $\Gamma$  act on the universal cover by deck transformations.

**Proof.** A nontrivial deck transformation must act freely. Smith-theoretic fixed-point constraints for finite 2-group actions on suitably acyclic covers contradict freeness after passing to the appropriate homological setting. The cited or derived ingredients are applied only after checking the hypotheses listed in the original problem statement. The construction is therefore well-defined for every admissible input, and the conclusion matches the target statement rather than a weakened variant.

**Boundary and consistency checks.** The proof covers the following edge cases explicitly:

- degenerate inputs allowed by the statement

The proof invokes the Smith-theory fixed-point theorem in the manifold category and checks the homological acyclicity hypothesis after passing to the relevant finite 2-subgroup. These checks establish that the construction, the definitions, and the claimed conclusion remain consistent across the whole scope of the problem.  $\square$